BT101
Introductory Biology
(Lecture 1)

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Course syllabus (BT 101, Introductory Biology)

- **Nutrients, bioenergetics and cell metabolism:** Essential nutrients to sustain life; biological energy and laws of thermodynamics, basics of aerobic and anaerobic glycolysis and citric acid cycle. **(Dr. Priyadarshi Satpati)**

- **Genes and chromosomes:** DNA, DNA replication; Central dogma of molecular biology: Transcription and translation; Mendelian Genetics; Genetic engineering/Cloning and its applications. **(Dr. Kusum K Singh)**

- **Biological systems:** Body systems required to sustain human physiology, special sense organs including hearing, taste, smell and visual receptors. **(Dr. Souptick Chanda)**

- **Evolution of life:** Origin of Life; Darwin's concepts of evolution; Biodiversity.

- **Cell, the structural and functional unit of life:** Three domains of life; cell types, cell organelles and structure; Basic biomolecules of cell. **(Dr. Rajkumar P Thummer)**

First Half

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Time line

Start date: 5th Jan, 2024 – 23rd Feb, 2024

Marks: 50

Mid-Sem : **26th February, 2024**

Instructors:

- Dr. Priyadarshi Satpati
- Dr. Kusum K Singh

Dr. Priyadarshi Satpati

Dr. Kususm K Singh

Second Half

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Date: 4th March 2024 – 26th April 2024

Marks : 50

End-Sem : 1st May, 2024

Instructors:

- Dr. Rajkumar P Thummer
- Dr. Souptick Chanda

Dr. Rajkumar P Thummer

Dr. Souptick Chanda

Grading = 100 marks = Mid-sem (50) + End-sem (50)

Text Books

1. J. L. Tymoczko, J. M. Berg and L. Stryer, Biochemistry, 8th Ed, W. H. Freeman & Co, 2015
2. D. L. Nelson and M. M. Cox, Lehninger Principles of Biochemistry, 7th Ed, Macmillan Worth, 2017.

Reference Books

1. N. Hopkins, J. W. Roberts, J. A. Steitz, J. Watson and A. M. Weiner, Molecular Biology of the Gene, 7th Ed, Benjamin Cummings, 1987.
2. C. R. Cantor and P. R. Schimmel, Biophysical Chemistry (Parts I, II and III), W.H. Freeman & Co., 1980.
3. C. C. Chatterjee, Human Physiology, Vol 1 & 2, 11th Ed, Medical Allied Agency, 1987.
4. Hall, B.K., Evolution: Principles and Processes, 1st Ed, Jones & Bartlett, 2011.

Course syllabus

Evolution of life:

- ❑ Origin of Life
- ❑ Darwin's concepts of evolution
- ❑ Biodiversity

Cell, the structural and functional unit of life:

- ❑ Three domains of life; cell types, cell organelles and structure
- ❑ Basic biomolecules of cell

Origin of life

- It is believed that life first emerged at least **3.8 billion (3.8×10^9) years ago**, approximately 750 million years after Earth was formed.

- How life originated and how the first cell came into being are matters of speculation, since these events cannot be reproduced in the laboratory.

- Nonetheless, several types of experiments provide important evidence bearing on some steps of the process.

Types of organisms

- ❑ **Prokaryotes**
- ❑ **Eukaryotes**

Prokaryon:

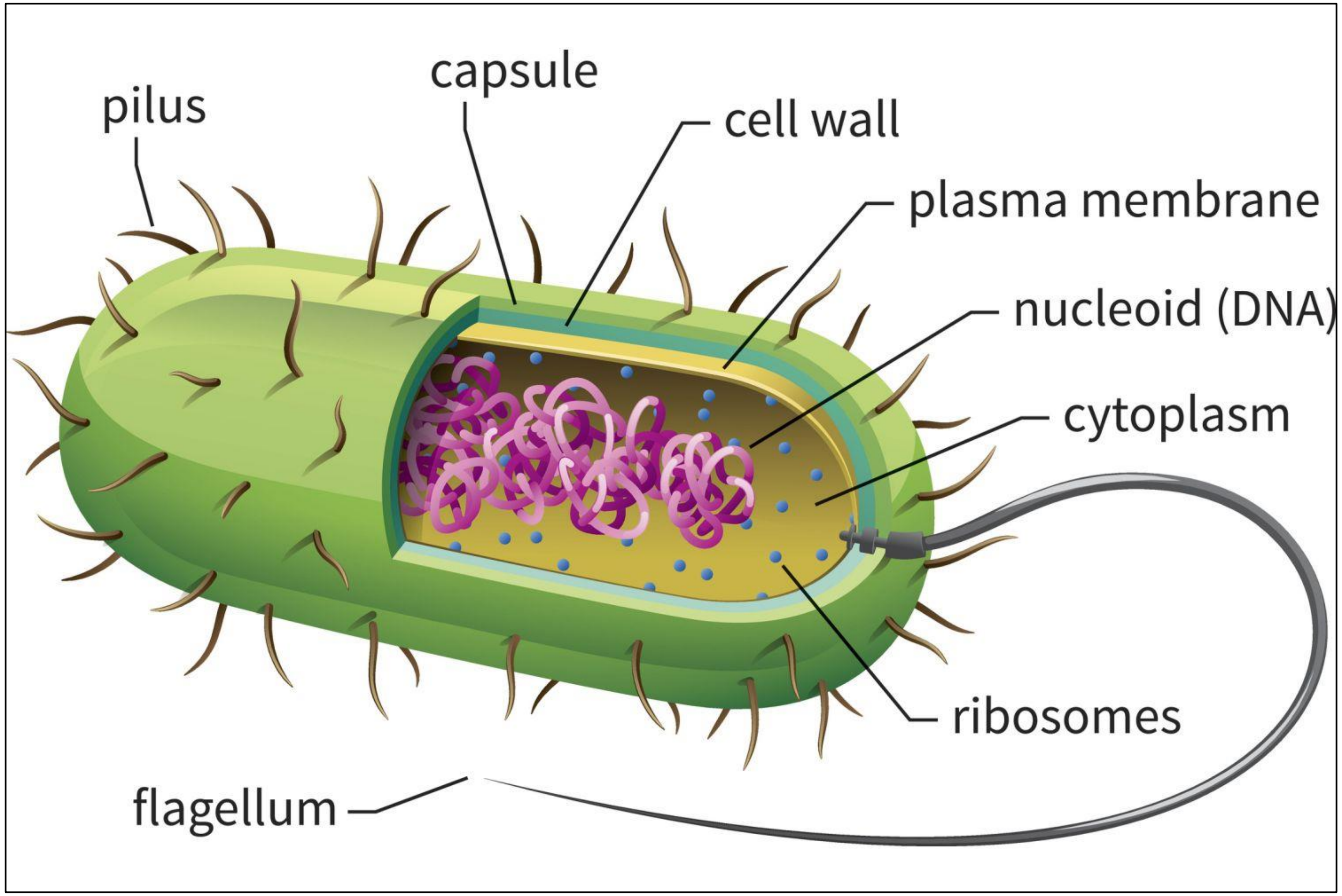
The Greek term ‘prokaryon’ is made up of two words, where ‘pro’ means ‘before’ and ‘karyon’ means ‘nucleus’. Thus ‘prokaryon’ means ‘primitive nucleus’.

Eukaryon:

The defining feature of eukaryotes is that their cells have nuclei. This gives them their name, from the Greek εὖ (eu, "well" or "good") and κάρυον (karyon, "nut" or "kernel", here meaning "nucleus").

Prokaryotes

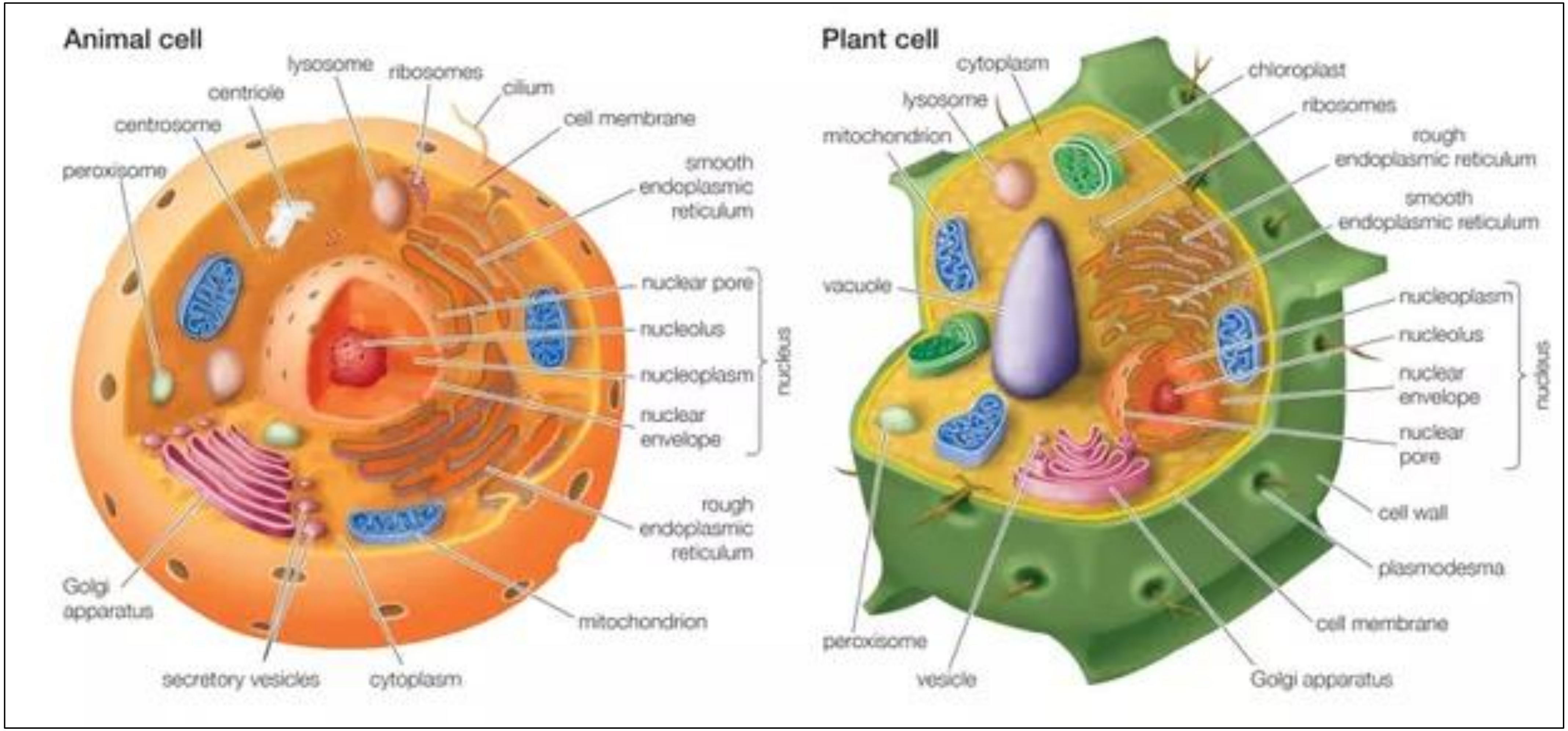
- **It is a single-celled organism that lacks a membrane-bound nucleus (karyon), mitochondria or any other membrane-bound organelle.**



(Structure of *Escherichia coli* cell, a Prokaryotic cell)

Eukaryotes

- It is any organism whose cells contain a nucleus and other organelles enclosed within membranes.



(Structure of a typical animal and plant cell; a Eukaryotic cell)

Thank you for your attention